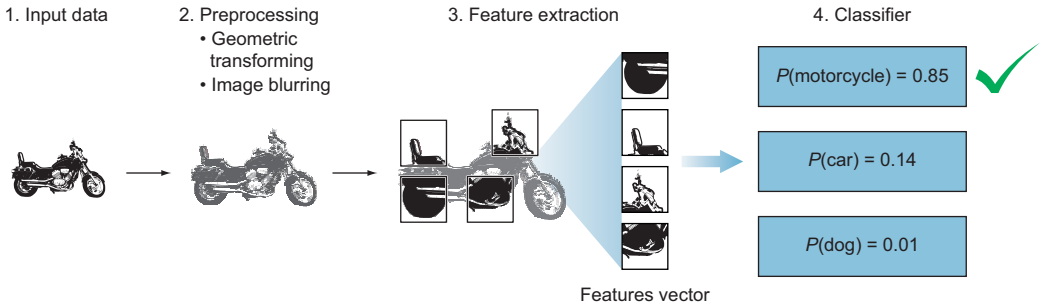

Figure 1.14 Using the machine learning model to predict the probability of the motorcycle object from the motorcycle, car, and dog classes

DEFINITIONS An *image classifier* is an algorithm that takes in an image as input and outputs a label or "class" that identifies that image. A *class* (also called a *category*) in machine learning is the output category of your data.

Here is how the image flows through the classification pipeline:

1. A computer receives visual input from an imaging device like a camera. This input is typically captured as an image or a sequence of images forming a video.
2. Each image is then sent through some preprocessing steps whose purpose is to standardize the images. Common preprocessing steps include resizing an image, blurring, rotating, changing its shape, or transforming the image from one color to another, such as from color to grayscale. Only by standardizing the images—for example, making them the same size—can you then compare them and further analyze them.
3. We extract features. *Features* are what help us define objects, and they are usually information about object shape or color. For example, some features that distinguish a motorcycle are the shape of the wheels, headlights, mudguards, and so on. The output of this process is a *feature vector* that is a list of unique shapes that identify the object.
4. The features are fed into a *classification model*. This step looks at the feature vector from the previous step and predicts the class of the image. Pretend that you are the classifier model for a few minutes, and let's go through the classification process. You look at the list of features in the feature vector one by one and try to determine what's in the image:
   a. First you see a *wheel* feature; could this be a car, a motorcycle, or a dog? Clearly it is not a dog, because dogs don't have wheels (at least, normal dogs, not robots). Then this could be an image of a car or a motorcycle.
   b. You move on to the next feature, the *headlights*. There is a higher probability that this is a motorcycle than a car.
   c. The next feature is *rear mudguards*—again, there is a higher probability that it is a motorcycle.